David Williams–Young, Ph.D.

Principal Quantum Architect
Microsoft Quantum
Microsoft Corporation
One Microsoft Way, Redmond, WA 98052

PROFESSIONAL INTERESTS

- Architecture and development of software for quantum computing applications.
- Quantum algorithms for chemistry and materials science, including state preparation and circuit synthesis.
- The intersection of high-performance computing and quantum information science.
- Development of high–performance electronic structure methods on emerging architectures.
- Development and microarchitecture optimization of high–performance linear algebra software.
- Leverage of modern C++ to develop clean, reusable, and extensible scientific software.

PROFESSIONAL EXPERIENCE

Principal Quantum Architect ..... September 2026 – Present
Principal Quantum Software Engineer ..... August 2024 – September 2026
Microsoft Quantum
Microsoft Corporation, Redmond, WA

- Engineering lead for quantum applications software at Microsoft Quantum, combining engineering management with hands-on technical leadership of product development.
- Lead architect and group lead for QDK/Chemistry and QDK/Analytics: set technical direction, software architecture, and engineering standards, and contribute directly to core development.
- Drive the integration of high-performance classical simulation and quantum algorithms into end-to-end quantum application workflows, translating research advances into production software.
- Previously technical lead for Accelerated DFT, a GPU-accelerated cloud service for density functional theory.

Research Scientist (Career) ..... March 2023 – August 2024
Research Scientist (Career-Track) ..... February 2021 – March 2023
Postdoctoral Fellow ..... July 2018 – February 2021
Applied Mathematics and Computational Research Division
Lawrence Berkeley National Laboratory, Berkeley, CA

Graduate Research Assistant ..... July 2013 – July 2018
Graduate Teaching Assistant ..... September 2013 – July 2018
Department of Chemistry
University of Washington, Seattle, WA

EDUCATION

Doctor of Philosophy (Ph.D.), Chemistry ..... May 2018
University of Washington, Seattle, WA
Adviser: Dr. Xiaosong Li
Dissertation: *Towards Efficient and Scalable Electronic Structure Methods for the Treatment of Relativistic Effects and Molecular Response*

Bachelor of Science (B.S., *Magna Cum Laude*), Chemistry, Mathematics ..... May 2013
Indiana University of Pennsylvania, Indiana, PA
Adviser: Dr. Jaeju Ko

COMPUTATIONAL PROFICIENCY

I am a specialist in the following computational areas:

- *Programming Languages:* C++, CUDA
- *Libraries / Paradigms:* OpenMP, MPI, BLAS/(Sca)LAPACK

I am very proficient in the following computational areas:

- *Programming Languages:* C, FORTRAN 77/95/03, HIP, SYCL
- *Scripting Languages:* Python, Julia, Octave, C/Bash Shell
- *Libraries / Paradigms:* TBB, OpenACC
- *Software:* Mathematica, MATLAB

I am the lead architect of the following software packages:

- **QDK/Chemistry**: Open-source toolkit for quantum chemistry applications on quantum computers (Microsoft)
- **QDK/Analytics**: Quantum algorithms for data analytics (Microsoft)

I am the primary developer of the following open-source software packages:

- **GauXC**: Enabling high-performance density functional theory calculations on exascale architectures
- **MACIS**: Massively parallel selected configuration interaction methods
- **ExchCXX**: GPU-accelerated library for the evaluation of exchange-correlation functionals
- **IntegratorXX**: Numerical quadrature for molecular integration grids

I have contributed to the development of the following software packages:

- Accelerated DFT (Microsoft)
- The Chronus Quantum (ChronusQ) Software Package
- NWChemEx
- Psi4
- Massively Parallel Quantum Chemistry (MPQC4)
- TiledArray
- Gaussian

PUBLICATIONS

46 in press publications, 5 preprints

* Indicates equal contribution to published work.

† Indicates that the publication was invited.

Preprints

5. Liu, D.; Zhu, S.; de Jong, W.A.; Fang, D.; **Williams–Young, D. B.**; Yang, C.; "*T-efficient unitary synthesis via block-sparse Hamiltonian simulation*" **2026**. arXiv:2610.18589.
4. Liu, D.; Wang, H.; Zhu, S.; Cong, J.; de Jong, W.A.; **Williams–Young, D. B.**; Yang, C.; "*Exponentially Compressed and Garbage-Free Alias Sampling for Polynomial State Preparation*" **2026**. arXiv:2610.03800.
3. Chen, Y.; Baker, N.A.; Gong, R.; Johnston, C.S.N.; Lackey, B.; Liu, H.; Schmidt, S.; Su, Y.; **Williams–Young, D. B.**; Yang, Y.; "*Clifford-efficient sparse state preparation for molecular wavefunctions*" **2026**. arXiv:2608.20593.
2. Baker, N.A.; Bilodeau, B.; Chen, C.; Chen, Y.; Eckhoff, M.; Efimovskaya, A.; Gasparotto, P.; van Gerwen, P.; Gong, R.; Hoang, K.; Hooshmand, Z.; Jenkins, A.J.; Johnston, C.S.N.; Li, R.R.; Liang, J.; Liu, H.; Mills, A.; Mörchen, M.; Nishibuchi, G.; Sun, C.; Ticehurst, B.; Troyer, M.; Unsleber, J.P.;

Wernli, S.; **Williams–Young, D. B.**; Zhang, B.; "*QDK/Chemistry: A Modular Toolkit for Quantum Chemistry Applications*" **2026**. arXiv:2601.15253.

1. Luise, G.; Huang, C.-W.; Vogels, T.; Kooi, D.P.; Ehlert, S.; Lanius, S.; Giesbertz, K.J.H.; Karton, A.; Gunceler, D.; Battaglia, S.; Simm, G.N.C.; Szabó, P.B.; Stanley, M.; Bruinsma, W.P.; Huang, L.; Wei, X.; Garrido Torres, J.; Katbashev, A.; Chavez Zavaleta, R.; Máté, B.; Kaba, S.-O.; Sordillo, R.; Chen, Y.; **Williams–Young, D. B.**; Bishop, C.M.; Hermann, J.; van den Berg, R.; Gori-Giorgi, P.; "*Accurate and scalable exchange-correlation with deep learning*" **2025**. arXiv:2506.14665.

Microsoft Quantum **2024-Present**

1. Ju, F.; Wei, X.; Huang, L.; Jenkins, A.J.; Xia, L.; Zhang, J.; Zhu, J.; Yang, H.; Shao, B.; Dai, P.; **Williams–Young, D. B.**; Mayya, A.; Hooshmand, Z.; Efimovskaya, A.; Baker, N.A.; Troyer, M.; Liu, H.; "*Acceleration without Disruption: DFT Software as a Service*" *J. Chem. Theory Comput.* **2024**. 20, 24, 10838–10851.

Lawrence Berkeley National Laboratory **2018-2024**

27. Shen, Y.; Camps, D.; Szasz, A.; Darbha, S.; Klymko, K.; **Williams–Young, D. B.**; Tubman, N.M.; Van Beeumen, R.; "*Estimating Eigenenergies from Quantum Dynamics: A Unified Noise-Resilient Measurement-Driven Approach*" *Quantum*; **2025**. 9, 1836.
26. Hirsbrunner, M.R.; Mullinax, J.W.; Shen, Y.; **Williams–Young, D. B.**; Klymko, K.; Van Beeumen, R.; Tubman, N.M.; "*A circuit-generated quantum subspace algorithm for the variational quantum eigensolver*" *J. Chem. Phys.* **2024**. 161, 164103.
25. Bylaska, E.; Panyala, A.; Bauman, N.; Peng, B.; Pathak, H.; Mejia-Rodriguez, D.; Govind, N.; **Williams–Young, D. B.**; Apra, E.; Bagusetty, A.; Mutlu, E.; Jackson, K.; Baruah, T.; Yamamoto, Y.; Pederson, M.; Withanage, K.; Pedroza-Montero, J.; Bilbrey, J.; Choudhury, S.; Firoz, J.; Herman, K.; Xantheas, S.; Rigor, P.; Vila, F.; Rehr, J.; Fung, M.; Grofe, A.; Johnston, C.; Baker, N.; Kaneko, K.; Liu, H.; Kowalski, K.; "*Electronic structure simulations in the cloud computing environment*" *J. Chem. Phys.* **2024**. 161, 150902.
24. Kovtun, M.; Lambros, E.; Liu, A.; Tang, D.; **Williams–Young, D. B.**; Li, X.; "*Accelerating Relativistic Exact-Two-Component Density Functional Theory Calculations with Graphical Processing Units*" *J. Chem. Theory Comput.* **2024**. 20, 18, 7694-7699
23. † Burton, H.; Dong, S.; Ghosh, S.; Gu, B.; Jackson, N.; Keefer, D.; Lu, Y.; Monroe, J.; Peng, B.; Pieri, E.; Spackman, P.; Vacher, M.; Vuckovic, S.; **Williams–Young, D. B.**; Yang, Z.; Yue, S.; Zerze, G.; Zhu, T.; "*Editorial: JCTC Early Career Board Selects*" *J. Chem. Theory Comput.*; **2024**. 20, 14, 5785–5787.
22. Poole, D.; **Williams–Young, D. B.**; Jiang, A.; Glick, Z.; Sherrill, C.D.; "*A modular, composite framework for the utilization of reduced-scaling Coulomb and Exchange construction algorithms: Design and implementation*"; *J. Chem. Phys.*; **2024**. 161, 052503.
21. Alvertis, A.M.; **Williams–Young, D. B.**; Bruneval, F.; Neaton, J.B.; "*Influence of Electronic Correlations on Electron–Phonon Interactions of Molecular Systems with the GW and Coupled Cluster Methods*" *J. Chem. Theory Comput.*; **2024**. 20, 14, 6175–6183.
20. † Blum, V.; Asahi, R.; Autschbach, J.; Bannwarth, C.; Bihlmayer, G.; Blügel, S.; Burns, L. A.; Crawford, T. D.; Dawson, W.; de Jong, W. A.; Draxl, C.; Filippi, C.; Genovese, L.; Giannozzi, P.; Govind, N.; Hammes-Schiffer, S.; Hammond, J. R.; Hourahine, B.; Jain, A.; Kanai, Y.; Kent, P. R. C.; Larsen, A. H.; Lehtola, S.; Li, X.; Lindh, R.; Maeda, S.; Makri, N.; Moussa, J.; Nakajima, T.; Nash, J. A.; Oliveira, M. J. T.; Patel, P. D.; Pizzi, G.; Pourtois, G.; Pritchard, B. P.; Rabani, E.; Reiher, M.; Reining, L.; Ren, X.; Rossi, M.; Schlegel, H. B.; Seriani, N.; Slipchenko, L. V.; Thom, A.; Valeev, E. F.; Van Troeye, B.; Visscher, L.; Vlcek, V.; Werner, H.-J.; **Williams–Young, D. B.**; Windus,

T.; "*Roadmap on methods and software for electronic structure based simulations in chemistry and materials*" *Electronic Structure*; **2024**. 6(4), 042501.

19. **Williams–Young, D. B.**; Yuwono, S.; DePrince III, A.E.; Yang, C.; "*Approximate Exponential Integrators for Time-Dependent Equation-of-Motion Coupled Cluster Theory*" *J. Chem. Theory Comput.*; **2023**. 19, 24, 9177–9186.

18. Di Felice, R.; Mayes, M.; Richard, R.; **Williams–Young, D. B.**; Chan, G.K.L; de Jong, W.A.; Govind, N.; Head-Gordon, M.; Hermes, M.; Kowalski, K.; Li, X.; Lischka, H.; Mueller, K.; Mutlu, E.; Niklasson, A.; Pederson, M.; Peng, B.; Shepard, R.; Valeev, E.; van Schilfgaarde, M.; Vlaisavljevich, B.; Windus, T.; Xantheas, S.; Zhang, X.; Zimmerman, P.; "*A Perspective on Sustainable Computational Chemistry Software Development and Integration*" *J. Chem. Theory Comput.*; **2023**. 19, 20, 7056–7076.

17. Ko, T.; Heindel, J.; Guan, X.; Head-Gordon, T.; **Williams–Young, D. B.**; Yang, C.; "*Using Diffusion Maps to Analyze Reaction Dynamics for a Hydrogen Combustion Benchmark Dataset*" *J. Chem. Theory Comput.*; **2023**. 19, 17, 5872–5885.

16. † **Williams–Young, D. B.**; Asadchev, A.; Popovici, D.T; Clark, D.; Waldrop, J.; Windus, T.L.; Valeev, E.F.; de Jong, W.A.; "*Distributed Memory, GPU Accelerated Fock Construction for Hybrid, Gaussian Basis Density Functional Theory*"; *J. Chem. Phys.*; **2023**. 158, 234104.

15. † **Williams–Young, D. B.**; Tubman, N.M.; Mejuto-Zaera, C.; de Jong, W.A.; "*A Parallel, Distributed Memory Implementation of the Adaptive Sampling Configuration Interaction Method*"; *J. Chem. Phys.*; **2023**. 158, 214109.

14. Richard, R.; Keipert, K.; Waldrop, J.; Keçeli, M.; **Williams–Young, D. B.**; Bair, R.; Boschen, J.; Crandall, Z.; Gasperich, K.; Mahmud, Q.; Panyala, A.; Valeev, E.F.; van Dam, H.; de Jong, W.A.; Windus, T.; "*PluginPlay: Enabling Exascale Scientific Software One Module at a Time*"; *J. Chem. Phys.*; **2023**. 158, 184801.

13. Shen, Y.; Klymko, K.; Sud, J.; **Williams–Young, D. B.**; de Jong, W.A.; Tubman, N.M; "*Real-Time Krylov Theory for Quantum Computing Algorithms*"; **2023**. *Quantum.* 7, 1066.

12. Gomes, N.; **Williams–Young, D. B.**; de Jong, W.A.; "*Computing the Many-Body Green's Function with Adaptive Variational Quantum Dynamics*"; *J. Chem. Theory Comput.*; **2023**. 19, 11, 3313–3323.

11. Sid-Lakhdar, W.M.; Cayrols, S.; Bielich, D.; Abdelfattah, A.; Luszczek, P.; Gates, M.; Tomov, S.; Johansen, H.; **Williams–Young, D. B.**; Davis, T.; Dongarra, J.; Anzt, H.; "*PAQR: Pivoting Avoiding QR factorization*"; 2023 IEEE International Parallel and Distributed Processing Symposium (IPDPS), St. Petersburg, FL, USA; **2023**, pp. 322-332.

10. Mejuto-Zaera, C.; Tzeli, D.; **Williams–Young, D. B.**; Tubman, N.M.; Matoušek, M.; Brabec, J.; Veis, L.; Xantheas, S.; de Jong, W.A.; "*The Effect of Geometry, Spin and Orbital Optimization in Achieving Accurate, Correlated Results for Iron-Sulfur Cubanes*"; *J. Chem. Theory Comput.*; **2022**. 18(2), 687–702.

9. Bez, J.L.; Tang, H.; Xie, B.; **Williams–Young, D. B.**; Latham, R.; Ross, R.; Oral, S.; Byna, S.; "*I/O Bottleneck Detection and Tuning: Connecting the Dots using Interactive Log Analysis*"; *2021 IEEE/ACM 6th International Parallel Data Systems Workshop (PDSW)*, **2021**.

8. **Williams–Young, D. B.**; Bagusetty, A.; de Jong, W.A.; Doerfler, D.; van Dam, H.J.J.; Vazquez–Mayagoitia, A.; Windus, T.L.; Yang, C.; "*Achieving Performance Portability in Gaussian Basis Set Density Functional Theory on Accelerator Based Architectures in NWChemEx*"; *Parallel Computing*, **2021**, 108, 102829.

7. Ahmed, H.; **Williams–Young, D. B.**; Ibrahim, K.Z.; Yang, C.; "*Performance Modeling and Tuning for DFT Calculations on Heterogeneous Architectures*"; *22nd IEEE International Workshop on Parallel and Distributed Scientific and Engineering Computing (PDSEC 2021)*, **2021**, pp. 714–722.

6. Kowalski, K.; Bair, R.; Bauman, N.P.; Boschen, J.S.; Bylaska, E.J.; Daily, J.; de Jong, W.A.; Dunning, T.; Govind, N.; Harrison, R.J.; Keceli, M.; Keipert, K.; Krishnamoorthy, S.; Kumar, S.; Mutlu, E.; Palmer, B.; Panyala, A.; Peng, B.; Richard, R.M.; Straatsma, T.P.; Sushko, P.; Valeev, E.F.; Valiev, M.; van Dam, H.J.J.; Waldrop, J.M.; **Williams–Young, D. B.**; Yang, C.; Zalewski, M.; Windus, T.L.;

"*From NWChem to NWChemEx: Evolving with the Computational Chemistry Landscape*"; *Chemical Reviews*, **2021**, 121(8), 4962–4998.

5. Yang, C.; Brabec, J.; Veis, L.; **Williams–Young, D. B.**; Kowalski, K.; "*Solving Coupled Cluster Equations by the Newton Krylov Method*"; *Frontiers in Chemistry*, **2020**, 8:590184.
4. † **Williams–Young, D. B.**; de Jong, W.A.; van Dam, H.J.J.; Yang, C.; "*On the Efficient Evaluation of the Exchange Correlation Potential on Graphics Processing Unit Clusters*"; *Frontiers in Chemistry*, **2020**, 8:581058.
3. **Williams–Young, D. B.**; Beckman, P.G.; Yang, C.; "*A Shift Selection Strategy for Parallel Shift-Invert Spectrum Slicing in Symmetric Self-Consistent Eigenvalue Computation*"; *ACM Trans. Math. Soft.*, **2020**, 46, 4, Article 35 (September 2020).
2. **Williams–Young, D. B.**; Yang, C.; "*Parallel Shift-Invert Spectrum Slicing on Distributed Architectures with GPU Accelerators*" in *Proceedings of the 49th International Conference on Parallel Processing (ICPP'20)*, **2020**.
1. Peng, B.; Van Beeumen, R.; **Williams–Young, D. B.**; Kowalski, K.; Yang, C.; "*Approximate Green's Function Coupled Cluster Method Employing Effective Dimension Reduction*"; *J. Chem. Theory Comput.*, **2019**, 15(5), 3185–3196.

University of Washington 2013-2018

18. Koulias, L.N.; **Williams–Young, D. B.**; Nascimento, D.R.; DePrince, A.E.; Li, X.; "*Relativistic Real-Time Time-Dependent Equation-of-Motion Coupled-Cluster*" *J. Chem. Theory Comput.*, **2019**, 15(12), 6617–6624.
17. Sun, S.; Beck, R.; **Williams–Young, D. B.**; Li, X.; "*Simulating Magnetic Circular Dichroism Spectra with Real-Time Time-Dependent Density Functional Theory in Gauge Including Atomic Orbitals*" *J. Chem. Theory Comput.*, **2019**, 15(12), 6824–6831.
16. † **Williams–Young, D. B.**; Petrone, A.; Sun, S.; Stetina, T.F.; Lestrange, P.; Hoyer, C.E.; Nascimento, D.R.; Koulias, L.; Wildman, A.; Kasper, J.; Goings, J.J.; Ding, F.; DePrince, A.E.; Valeev, E.F.; Li, X.; "*The Chronus Quantum (ChronusQ) Software Package*" *WIREs Comput. Mol. Sci.* **2020**, 10(2), e1436.
15. Stetina, T.F.; Sun, S.; **Williams–Young, D. B.**; Li, X.; "*Modeling Magneto-Photoabsorption Using Time-Dependent Complex Generalized Hartree-Fock*" *ChemPhotoChem*, **2019**, 3(9), 739–746.
14. Hoyer, C.; **Williams–Young, D. B.**; Huang, C.; Li, X.; "*Embedding Non-Collinear Two-Component Electronic Structure in a Collinear Quantum Environment*" *J. Chem. Phys.*, **2019**, 150(17), 174114.
13. Sun, S.; **Williams–Young, D. B.**; Li, X.; "*An Ab Initio Linear Response Method for Computing Magnetic Circular Dichroism Spectra with Non-Perturbative Treatment of Magnetic Field*"; *J. Chem. Theory Comput.*, **2019**, 15(5), 3162–3169.
12. Sun, S.; **Williams–Young, D. B.**; Stetina, T.F.; Li, X.; "*Generalized Hartree-Fock with a Non-perturbative Treatment of Strong Magnetic Fields: Application to Molecular Spin Phase Transitions*"; *J. Chem. Theory Comput.*, **2019**, 15(1), 348–356.
11. Petrone, A.*; **Williams–Young, D. B.***; Sun, S.; Stetina, T. F.; Li, X.; "*An Efficient Implementation of Two-Component Relativistic Density Functional Theory with Torque-Free Auxiliary Variables*"; *Eur. Phys. J. B*, **2018**, 91(7), 169.
10. Kasper, J.; **Williams–Young, D. B.**; Vecharynski, E.; Yang, C.; Li, X.; "*A Well-Tempered Hybrid Method for Solving Challenging Time-Dependent Density Functional Theory (TDDFT) Systems*"; *J. Chem. Theory Comput.*, **2018**, 14(4), 2034–2041.
9. Lestrange, P.; **Williams–Young, D. B.**; Petrone, A.; Jiménez–Hoyos, C.; Li, X.; "*Efficient Implementation of Variation After Projection Generalized Hartree–Fock*" *J. Chem. Theory Comput.*, **2018**, 14(2), 588–596.

8. Barclay, M. S.; Quincy, T. J.; **Williams–Young, D. B.**; Caricato, M.; Elles, C. G.; " *Accurate Assignments of Excited-State Resonance Raman Spectra: A Benchmark Study Combining Experiment and Theory*" *J. Phys. Chem. A.*, **2017**, 121(41), 7937–7946.
7. Van Beeumen, R.; **Williams–Young, D. B.**; Kasper, J.; Yang, C.; Ng, E. G.; Li, X.; "*Model Order Reduction Algorithm for Estimating the Absorption Spectrum*" *J. Chem. Theory Comput.*, **2017**, 13(10), 4950–4961.
6. Egidi, F.*; **Williams–Young, D. B.***; Baiardi, A.*; Bloino, J.; Scalmani, G.; Frisch, M.; Li, X.; Barone, V.; "*Effective Inclusion of Mechanical and Electrical Anharmonicity in Excited Electronic States: the VPT2-TDDFT Route*" *J. Chem. Theory Comput.*, **2017**, 13(6), 2789–2803.
5. Petrone, A.*; **Williams–Young, D. B.***; Lingerfelt, D. B.; Li, X.; "*Ab Initio Excited-State Transient Raman Analysis*" *J. Phys. Chem. A.*, **2017**, 121(20), 3958–3965.
4. Petrone, A.; Lingerfelt, D. B.; **Williams–Young, D. B.**; Li, X.; "*Ab Initio Transient Vibrational Spectral Analysis*" *J. Phys. Chem. Lett.*, **2016**, 7, 4501–4508.
3. **Williams–Young, D. B.**; Goings, J.; Li, X.; "*Accelerating Real–Time Time-Dependent Density Functional Theory with a Non–Recursive Chebyshev Expansion of the Quantum Propagator*" *J. Chem. Theory Comput.*, **2016**, 12(11) 5333–5338.
2. **Williams–Young, D. B.**; Egidi, F.; Li, X.; "*Relativistic Two-Component Particle-Particle Tamm–Dancoff Approximation*" *J. Chem. Theory Comput.*, **2016**, 12(11), 5379–5384.
1. Lingerfelt, D. B.; **Williams–Young, D. B.**; Petrone, A; Li, X.; "*Direct ab Initio (Meta-)Surface-Hopping Dynamics*", *J. Chem. Theory Comput.*, **2016**, 12(3), 935–945.

HONORS

CCG Excellence Award for Graduate Students — The Chemical Computing Group (2017)
MolSSI Software Fellow — Molecular Sciences Software Institute (2017-2018)
Lloyd E. and Florence M. West Fellowship in Chemistry — Lloyd E. and Florence M. West (2016)
Excellence in Chemistry Graduate Fellowship Award (ECGFA) — University of Washington (2013)

PROFESSIONAL SERVICE

Leadership Activities

Journal of Chemical Theory and Computation Early Career Board (2024)
NWChemEx Executive Board (2023-2024)

Journal Review

Molecular Physics, *The Journal of Chemical Physics*, *The International Journal of Quantum Chemistry*, *Computer Physics Communications*, *The Journal of Chemical Theory and Computation*, *Journal of Computational Physics*, *The Journal of Physical Chemistry*, *The Journal of Physical Chemistry A*

Proposal Review

NSF Office of Advanced Cyberinfrastructure, *DOE Office of Science, Office of Basic Energy Sciences*, *Natural Sciences and Engineering Research Council of Canada*

Conference Symposium Organization

SIAM Conference on Computational Science and Engineering, *SIAM Conference on Parallel Processing for Scientific Computing*

PRESENTATIONS

* Indicates an invited presentation

22. * **Williams–Young, D. B.**; "*Quantum, AI, HPC and the Path to Commercial Advantage*"; HPC/AI Integration with Quantum Computing (HAIQ) Workshop, International Symposium on High-Performance Computer Architecture (HPCA); Las Vegas, NV; **2025**; Oral Presentation.
21. * **Williams–Young, D. B.**; "*Pushing Chemistry Boundaries with Quantum and High-Performance Computing*"; International Workshop on Massively Parallel Programming for Quantum Chemistry and Physics (MPQCP 2025); Kobe, Japan; **2025**; Oral Presentation.
20. * **Williams–Young, D. B.**; "*Massively Parallel Selected Configuration Interaction for First-Principles Quantum Chemistry Applications*"; Society for Industrial and Applied Mathematics Conference on Parallel Processing for Scientific Computing (SIAM-PP24); Baltimore, MD; **2024**; Oral Presentation.
19. **Williams–Young, D. B.**; "*Revisiting Pseudospectral Methods for Hybrid Kohn-Sham Density Functional Theory in the Age of Exascale Computing*"; American Physical Society March Meeting 2023; Las Vegas, NV; **2023**; Oral Presentation.
18. * **Williams–Young, D. B.**; "*Perspective on High-Performance Algorithms for Eigenvalue Problems in Physical Simulations*"; Society for Industrial and Applied Mathematics Conference on Computational Science and Engineering (SIAM-CSE23); Amsterdam, The Netherlands; **2023**; Oral Presentation.
17. * **Williams–Young, D. B.**; "*Modular Design Patterns for the Sustainable Development of High-Performance Computational Chemistry Software*"; Basic Energy Sciences Workshop on Sustainable Computational Chemistry Software Development, Integration, and Application; Seattle, WA; **2022**; Oral Presentation.
16. * **Williams–Young, D. B.**; "*Low-Order, Scalable Eigensolvers for Electronic Structure Simulations Based on Spectrum Slicing*"; CECAM Workshop on Challenges and Advances in Solving Eigenproblems for Electronic Structure Theory; Lausanne, Switzerland; **2022**; Oral Presentation.
15. **Williams–Young, D. B.**; "*Revisiting Pseudospectral Methods for Hybrid Kohn-Sham Density Functional Theory in the Age of Exascale Computing*"; 263rd American Chemical Society National Meeting & Exposition; San Diego, CA; **2022**; Oral Presentation.
14. * **Williams–Young, D. B.**; "*Revisiting Pseudospectral Methods for Hybrid Kohn-Sham Density Functional Theory in the Age of Exascale Computing*"; Society for Industrial and Applied Mathematics Conference on Parallel Processing for Scientific Computing (SIAM-PP22); Virtual; **2022**; Oral Presentation.
13. **Williams–Young, D. B.**; "*Developing (Semi)numerical Methods for Hybrid Kohn-Sham Density Functional Theory in the Age of Exascale Computing*"; The International Chemical Congress of Pacific Basin Societies (Pacifichem '21); Virtual; **2021**; Oral Presentation.
12. * **Williams–Young, D. B.**; "*Parallel Shift-Invert Spectrum Slicing for Symmetric Self-Consistent Eigenvalue Computation*"; Society for Industrial and Applied Mathematics Conference on Computational Science and Engineering (SIAM-CSE21); Virtual; **2021**; Oral Presentation.
11. **Williams–Young, D. B.**; "*Parallel Shift-Invert Spectrum Slicing on Computing Clusters with GPU Accelerators*"; 49th International Conference on Parallel Processing (ICPP20); Virtual; **2020**; Oral Presentation.
10. **Williams–Young, D. B.**; Yang, C.; "*Parallel Shift-Invert Spectrum Slicing for Symmetric Self-Consistent Eigenvalue Computation*"; Society for Industrial and Applied Mathematics Conference on Parallel Processing for Scientific Computing (SIAM-PP20); Seattle, WA; **2020**; Oral Presentation.
9. **Williams–Young, D. B.**; Van Beeumen, R.; Yang, C.; Li, X.; "*A General Model Order Reduction Scheme for the Evaluation of Spectroscopic Properties and Excited States*"; American Physical Society March Meeting 2019; Boston, MA; **2019**; Oral Presentation.

8. * **Williams–Young, D. B.**; "*On the High Performance Implementation of Quaternionic Matrix Operations*"; Society for Industrial and Applied Mathematics Conference on Computational Science and Engineering (SIAM-CSE19); Spokane, WA; **2019**; Oral Presentation.
7. **Williams–Young, D. B.**; Van Beeumen, R.; Yang, C.; Li, X.; "*A Novel Model Reduction Algorithm for the Efficient Evaluation of Molecular Response Properties*"; 254th American Chemical Society National Meeting & Exposition; Washington, DC; **2017**; Poster.
6. * **Williams–Young, D. B.**; "*Studying Semi–Classical Molecular Light–Matter Interaction through Time–Dependent Density Function Theory*"; High Performance Computing Seminar; University of Washington, Seattle, WA; **2017**; Oral Presentation.
5. **Williams–Young, D. B.**; Goings, J.J.; Li, X.; "*Accelerating Real–Time Time–Dependent Density Functional Theory with a Chebyshev Expansion of the Quantum Propagator*"; Theory and Applications of Computational Chemistry (TACC) 2016; Seattle, WA; **2016**; Poster.
4. **Williams–Young, D. B.**; Yang, W.; Li, X.; "*Moving past the particle-hole description of excited states: Affordable methodologies*"; The International Chemical Congress of Pacific Basin Societies (Pacifichem '15); Honolulu, HI; **2015**; Oral Presentation.
3. **Williams–Young, D. B.**; Ko, J.; Ondrechen, M.J.; "*Computational approach to the prediction of enzyme specificities*"; 245th American Chemical Society National Meeting & Exposition; New Orleans, LA; **2013**; Poster.
2. **Williams–Young, D. B.**; Ko, J.; "*Prediction of Relative Activities of Enzymes using Computed Chemical Properties*"; American Chemical Society Student Member Symposium; Duquesne University, Duquesne, PA; **2012**; Poster.
1. **Williams–Young, D. B.**; Ko, J.; "*Prediction of Relative Activities of Enzymes using Computed Chemical Properties*"; Undergraduate Research Forum; Indiana University of Pennsylvania. Indiana, PA; **2012**; Poster.

MEMBERSHIPS

Alpha Chi Sigma (AXΣ), ΓT Chapter, Professional Chemistry Fraternity
American Chemical Society (ACS), Computers in Chemistry Division
American Physical Society (APS)
Society for Industrial and Applied Mathematics (SIAM), Activity Group on Computational Science and Engineering